

Group 38 Flight Data Analysis Project Report

This group was comprised of six CSLL students – Emily Eulitz, Shane Jones, Autumn Kaiser, Lauren McGaughey, Brynne Mittleider, and Ethan Ó Mórdha.

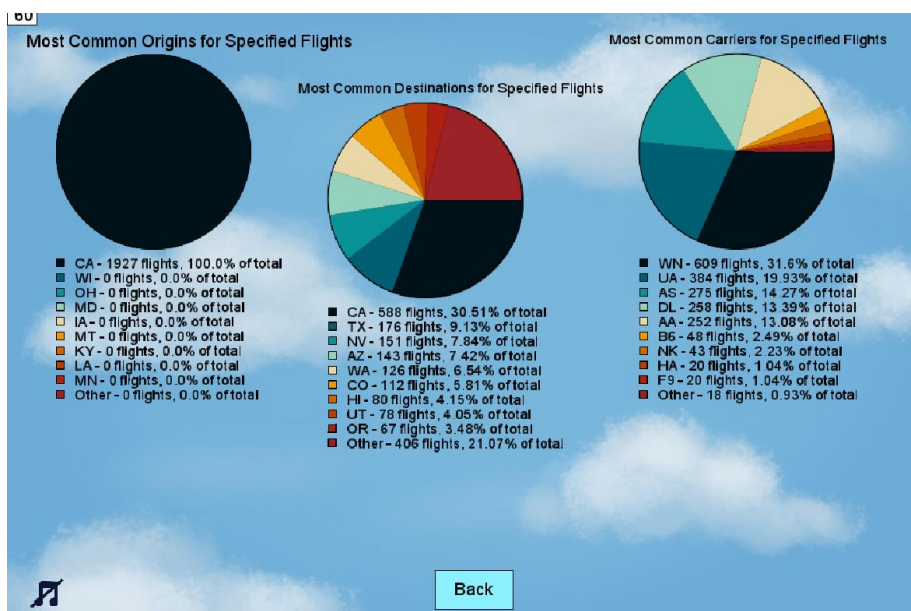
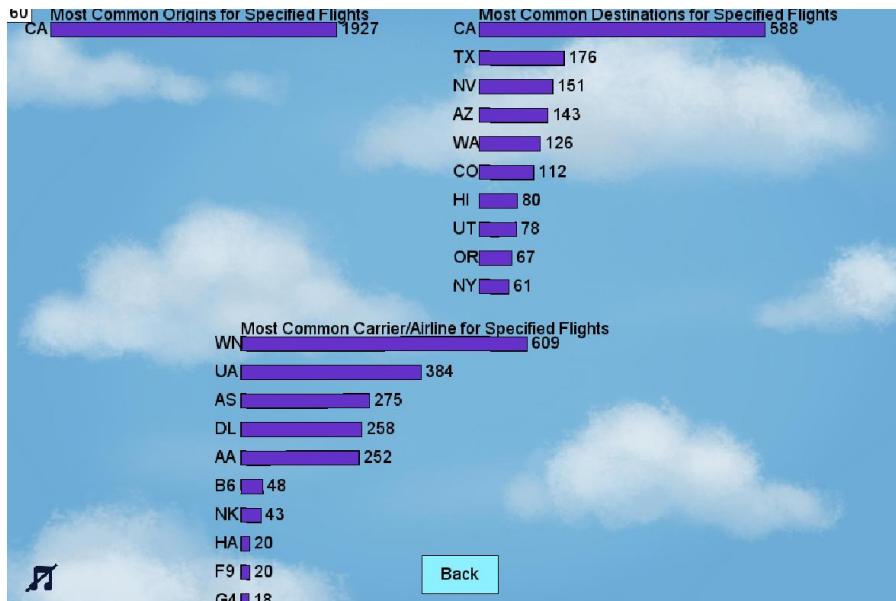
The objective of this project was to create an application to enable one to efficiently search through, and explore, data about domestic flights in the U.S. This application should allow users to search by a variety of criteria quickly, with an easy-to-use interface. By utilising pie charts and bar charts, as well as a map of the U.S., this project presents easily digestible visual data to users; Users can also gather data based on their own unique search specifications.

Throughout the creation of our site, we made use of websites like Processing and W3Schools to get a baseline understanding of different areas of code and to discover how to best apply our knowledge to the functionality of the project. We referred to some of the examples in Professor Murala's lecture slides when refreshing ourselves on Processing code. To add music to our application, we made use of the Processing sound library. Our flight data came from the U.S. Bureau of Transportation Statistics website.

Our site consists of a Welcome screen, Search screen, and a Results screen from which you can access a Map screen, Bar Chart screen, and Pie Chart screen. From the Search screen, one can search for flights that match any number of criteria, including by airline, departure state, arrival state, a range of departure/arrival times, a range of dates, and whether the flight was late or cancelled. Once you complete your search and see your results, you can sort them by several factors depending on what is most relevant to your interests, by simply ticking a box. By selecting a singular flight from the list of matching results, you are taken to the Map screen which shows the flight path between states. Both the Bar and Pie Chart screens will show statistics based on the specifications of your search. At any point, you can return to the search screen and edit your search criteria to whatever you'd like to see next.

In the first week of working on the project, the group met to discuss and collaborate on the outline of the visuals and code for our application. Autumn and Ethan headed up UX/UI design. Autumn created some mock-ups of how many screens there should be, what should be on them, and their layout. She also created art assets for the

background images and buttons. Brynne, our project manager, kept meeting notes each week in a Google Doc used as reference by the group, and created a guide to using GitHub for us. Brynne converted the .csv file to a Table class, which was then converted into an ArrayList of Flights by Emily (who wrote the Flight class), for ease of reference and file analysis. Ethan added an fps counter early on to give visual indication of slowdown. Brynne made the Input class, its subclass Dropdown, and the DropdownItem class, for user search input. Lauren created two-way sliders, a calendar from which a range of dates could be selected, and tickboxes, for additional search filtration. After some group problem-solving, Emily created the core backend functions for filtering data based on many possible criteria set by the user. She also wrote a clear guide to using this function, and others she'd written, including `percentWhichMatchThisCriterion()`, `mostPopValue()`, and functions to sort data by ranges of dates/times, so the group could utilize these functions effectively. Brynne created the `filteredData()` function which takes the input from their Dropdowns and Lauren's filters and uses Emily's Data class to identify all flights matching the input. At this point, we had achieved our first goal of being able to search through the full file based on a number of user specifications. Shane wrote and implemented a Screen class, and a Widget class, which contained three constructors, enabling us to create non-clickable or clickable text widgets, and clickable image widgets. He also implemented the functionality to switch screens with widgets. On the Results screen, Brynne visualized the data in a scrollable table using their Sheet and Scrollbar classes, and made the data sortable using the `sortSheet()` function. Ethan worked on a Map screen, with a map of the U.S. that he coded to display individual flight paths. Brynne made the individual flights of the Sheet clickable in order to send their data to this screen. Autumn continued work on design and formatting, and made a Plane class to add flying planes across the home screen. She also added music, and worked with Shane to create a music mute/unmute button. Lauren and Shane also worked on fixing some newer formatting issues with the Search screen. Brynne and Emily collaborated on coding functional bar charts for our data screens, and Autumn and Emily implemented pie charts that pulled from the data as well. Here is an example of data we presented in graphs in our finished project:



Towards the end of the project, we all contributed to neatening the aesthetics and formatting of our site, and debugging the code. From user testing with a small sample size of family and friends, some of whom with backgrounds in tech and some without, we found our site to be intuitive, functional, efficient, and overall, a success.

This project provides users a multitude of options to empower their search, including criteria as specific as selecting a range of dates or arrival/departure times. Our intuitive UX/UI design, our easily accessible visual data, and effective data scanning methods, ensure a smooth and efficient experience for the user. Ultimately, through this methodology, this application accomplishes our goal of enabling ourselves, and users, to discover, analyse, and deeply explore a vast quantity of flight data.

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